

Semaglutide for chronic kidney disease in type 2 diabetes

Critical appraisal using CASP points

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Vlado Perkovic, M.B., B.S., Ph.D., Katherine R. Tuttle, M.D., Peter Rossing, M.D., D.M.Sc., Kenneth W. Mahaffey, M.D., Johannes F.E. Mann, M.D., George Bakris, M.D., Florian M.M. Baeres, M.D., Thomas Idorn, M.D., Ph.D., Heidrun Bosch-Traberg, M.D., Nanna Leonora Lausvig, M.Sc., and Richard Pratley, M.D., for the FLOW Trial Committees and Investigators*

ABSTRACT

BACKGROUND

Patients with type 2 diabetes and chronic kidney disease are at high risk for kidney failure, cardiovascular events, and death. Whether treatment with semaglutide would mitigate these risks is unknown.

METHODS

We randomly assigned patients with type 2 diabetes and chronic kidney disease (defined by an estimated glomerular filtration rate [eGFR] of 50 to 75 ml per minute per 1.73 m² of body-surface area and a urinary albumin-to-creatinine ratio [with albumin measured in milligrams and creatinine measured in grams] of >300 and <5000 or an eGFR of 25 to <50 ml per minute per 1.73 m² and a urinary albumin-to-creatinine ratio of >100 and <5000) to receive subcutaneous semaglutide at a dose of 1.0 mg weekly or placebo. The primary outcome was major kidney disease events, a composite of the onset of kidney failure (dialysis, transplantation, or an eGFR of <15 ml per minute per 1.73 m²), at least a 50% reduction in the eGFR from baseline, or death from kidney-related or cardiovascular causes. Prespecified confirmatory secondary outcomes were tested hierarchically.

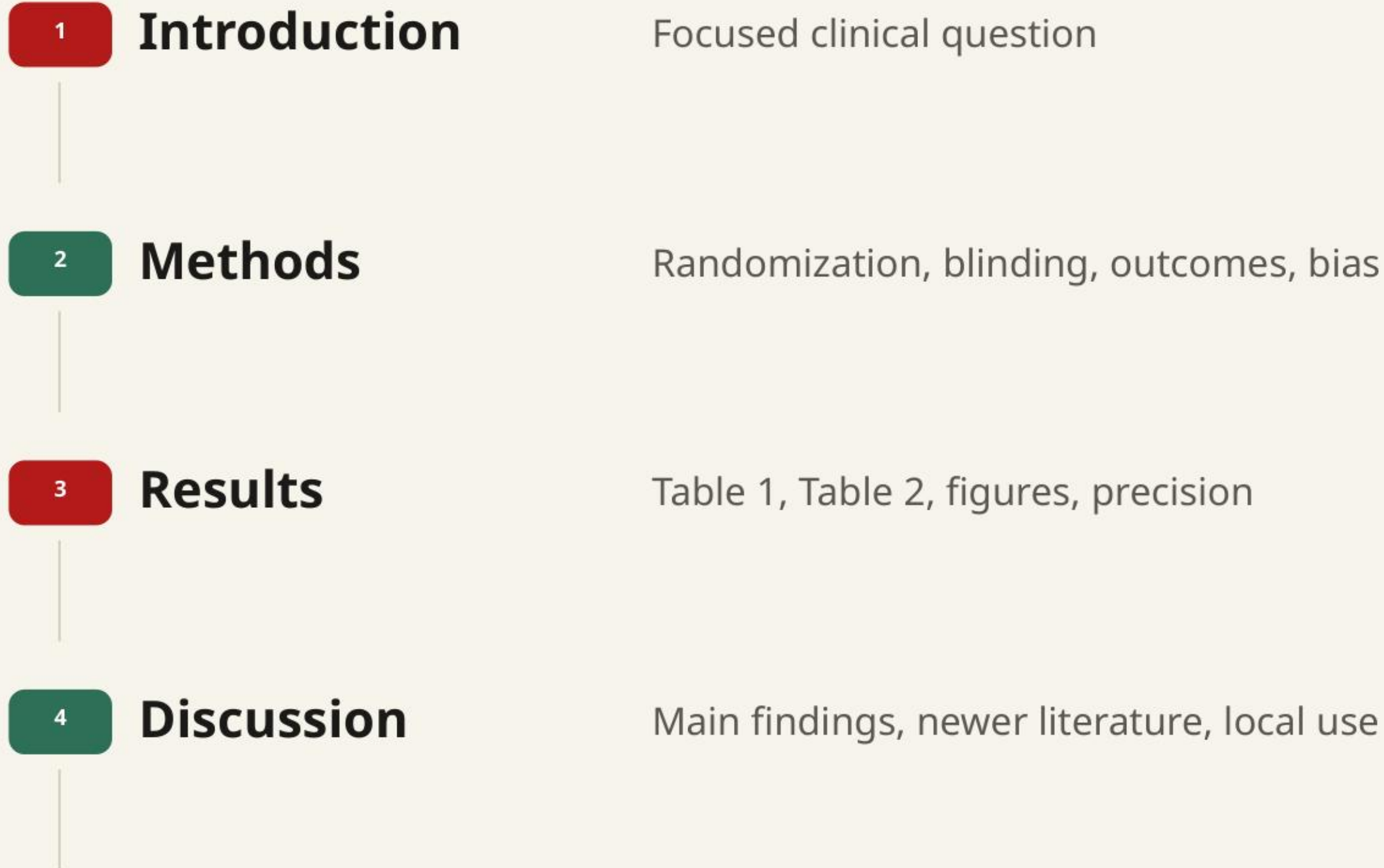
RESULTS

Among the 3533 participants who underwent randomization (1767 in the semaglutide group and 1766 in the placebo group), median follow-up was 3.4 years, after early trial cessation was recommended at a prespecified interim analysis. The risk of a primary-outcome event was 24% lower in the semaglutide group than in the placebo

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How this appraisal is organized

Same structure as the paper, with CASP checkpoints after each section.



INTRODUCTION

Why this trial mattered

Residual kidney and cardiovascular risk remained high despite RAS blockade, SGLT2 inhibitors and finerenone.

Clinical problem: T2D + CKD patients are at high risk of kidney failure, cardiovascular events and death.

Evidence gap: before FLOW, GLP-1 RA kidney data were mostly secondary or post hoc outcomes, not a dedicated kidney-outcome trial.

Research question: Does once-weekly subcutaneous semaglutide 1.0 mg reduce major kidney disease events compared with placebo?

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PICO summary

A clean, clinically focused randomized trial question.

- P** — Adults with T2D, HbA1c $\leq 10\%$, high-risk CKD, mostly on ACEi/ARB
- I** — Semaglutide 1.0 mg subcutaneous once weekly
- C** — Matching placebo
- O** — Major kidney disease events: kidney failure, $\geq 50\%$ eGFR fall, kidney/CV death

CASP: Did the trial address a clearly focused issue?

Introduction appraisal

Strengths

- Clear target population: T2D with high-risk CKD.
- Clinically important endpoint, not just albuminuria.
- Addresses an unmet question for GLP-1 receptor agonists.

Concerns / context

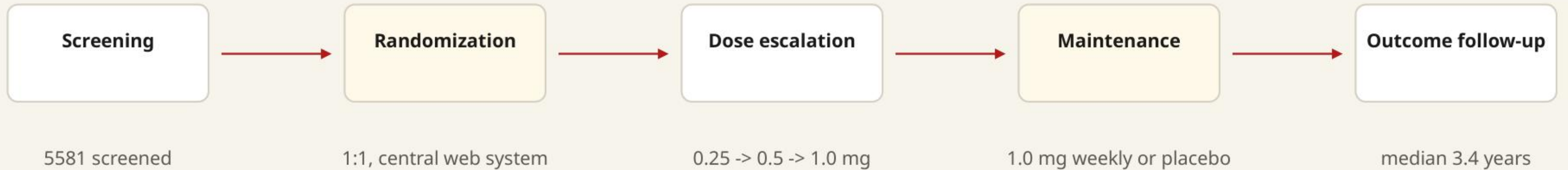
- Sponsor interest is high because semaglutide is a commercial drug.
- Study population is high-risk; lower-risk CKD patients are not directly answered.
- The question is efficacy, not cost-effectiveness or access.

Verdict: Yes - clearly focused and clinically important.

METHODS

Trial design

International, randomized, double-blind, placebo-controlled, event-driven trial.



- 387 sites across 28 countries.
- Baseline stratification by SGLT2 inhibitor use.
- Trial stopped early after prespecified interim analysis for efficacy.
- Sponsor managed operations; independent verification of original data reported.

Eligibility and baseline therapy

Enriched for high-risk diabetic kidney disease.

Included

- Adults with type 2 diabetes; HbA1c $\leq 10\%$.
- eGFR 50-75 with UACR >300 to <5000 mg/g.
- or eGFR 25- <50 with UACR >100 to <5000 mg/g.
- Stable maximal/tolerated ACE inhibitor or ARB unless not tolerated.

Allowed background care

- SGLT2 inhibitors allowed and used for stratification.
- Mineralocorticoid receptor antagonists allowed.
- Insulin and lipid-lowering therapy common at baseline.
- This was add-on therapy, not replacement of standard care.

Outcomes and analysis

Clinically important outcomes with hierarchy to protect against false-positive secondary claims.

854

planned primary events for 90% power

ITT

all randomized participants analyzed

0.0322

updated two-sided significance threshold

3

confirmatory secondary outcomes

- Primary outcome: kidney failure, sustained $\geq 50\%$ eGFR decline, kidney-related death or cardiovascular death.
- Secondary hierarchy: total eGFR slope $\rightarrow$ major cardiovascular events $\rightarrow$ all-cause death.
- Blinded adjudication for non-laboratory outcomes.
- Supportive/exploratory outcomes were not adjusted for multiplicity.

CASP: Methods appraisal

Randomization, blinding, follow-up and outcome validity.

Randomized?	Yes - central web-based 1:1 allocation.
All accounted?	Mostly yes - vital status confirmed in 98.6%; 34 withdrew consent.
Blinded?	Yes - double-blind placebo-controlled; events adjudicated blinded.
Equal care?	Mostly yes - same follow-up design; background therapies allowed.
Important outcomes?	Yes - kidney failure, eGFR decline, CV events, death, safety.

Methodological cautions

Important when applying FLOW beyond the trial population.

- Early stopping can overestimate treatment effect, especially for later/harder outcomes.
- SGLT2 inhibitor use at baseline was only 15.6%, lower than current ideal CKD care.
- Finerenone/nonsteroidal MRA use was also limited because approvals evolved after trial start.
- Subgroups, including SGLT2 users, were underpowered for firm interaction conclusions.
- Most participants were White; Indian/South Asian representation was not reported separately.

Critical reading point: FLOW answers whether semaglutide works vs placebo on background care at the time of the study. It does not fully answer the best sequencing or combination strategy in 2026 practice.

RESULTS

Baseline characteristics: key takeaways from Table 1

Groups were clinically well balanced at randomization.

3533

randomized participants

66.6 yr

mean age

47.0

mean eGFR ml/min/1.73 m²

568

median UACR mg/g

15.6%

baseline SGLT2 inhibitor use

- Around 68.5% had A3 macroalbuminuria.
- 61.4% were on insulin; 80.2% were on lipid-lowering treatment.
- Female participants were 30.3%; Asian participants 23.9%.
- Mean HbA1c was 7.8%; mean BMI 32.0.

Table 1: baseline characteristics

Full baseline table from the paper.

Table 1. Characteristics of the Participants at Baseline.*			
Characteristic	Semaglutide (N = 1767)	Placebo (N = 1766)	Total (N = 3533)
Age — yr	66.6±9.0	66.7±9.0	66.6±9.0
Female sex — no. (%)	519 (29.4)	550 (31.1)	1069 (30.3)
Geographic region — no. (%)			
Asia	478 (27.1)	434 (24.6)	912 (25.8)
Europe	472 (26.7)	491 (27.8)	963 (27.3)
North America	423 (23.9)	442 (25.0)	865 (24.5)
Other	394 (22.3)	399 (22.6)	793 (22.4)
Race or ethnic group — no. (%)†			
White	1155 (65.4)	1168 (66.1)	2323 (65.8)
Asian	439 (24.8)	407 (23.0)	846 (23.9)
Black	78 (4.4)	82 (4.6)	160 (4.5)
Other	95 (5.4)	109 (6.2)	204 (5.8)
Hispanic or Latinx ethnic group — no. (%)†			
Yes	273 (15.4)	283 (16.0)	556 (15.7)
No	1421 (80.4)	1411 (79.9)	2832 (80.2)
Not reported	73 (4.1)	72 (4.1)	145 (4.1)
Glycated hemoglobin level — %	7.8±1.3	7.8±1.3	7.8±1.3
Body-mass index‡	31.9±6.1	32.0±6.5	32.0±6.3
Body weight — kg	89.5±19.8	89.8±21.2	89.6±20.5
Systolic blood pressure — mm Hg	138.9±16.1	138.4±15.4	138.6±15.8
Diastolic blood pressure — mm Hg	76.8±10.0	76.1±10.0	76.4±10.0
Diabetes duration — no. (%)			
<15 yr	774 (43.8)	753 (42.6)	1527 (43.2)
≥15 yr	992 (56.1)	1013 (57.4)	2005 (56.8)
Previous myocardial infarction or stroke — no. (%)	405 (22.9)	403 (22.8)	808 (22.9)
Chronic heart failure — no. (%)	342 (19.4)	336 (19.0)	678 (19.2)
Smoking status — no. (%)§			
Current smoker	223 (12.6)	206 (11.7)	429 (12.1)
Previous smoker	661 (37.4)	696 (39.4)	1357 (38.4)
Never smoked	883 (50.0)	864 (48.9)	1747 (49.4)
eGFR — ml/min/1.73 m²¶	46.9±15.6	47.1±14.7	47.0±15.2
eGFR distribution — no. (%)¶			
≥60 ml/min/1.73 m²	366 (20.7)	353 (20.0)	719 (20.4)
≥45 to <60 ml/min/1.73 m²	515 (29.1)	540 (30.6)	1055 (29.9)
≥30 to <45 ml/min/1.73 m²	667 (37.7)	691 (39.1)	1358 (38.4)
<30 ml/min/1.73 m²	218 (12.3)	182 (10.3)	400 (11.3)
Median urinary albumin-to-creatinine ratio	582.3	557.8	567.6
Category of albuminuria — no. (%)**			
A1, normoalbuminuria	52 (2.9)	57 (3.2)	109 (3.1)
A2, microalbuminuria	509 (28.8)	495 (28.0)	1004 (28.4)
A3, macroalbuminuria	1205 (68.2)	1214 (68.7)	2419 (68.5)

Table 1 continued: baseline medication use

Background therapy matters for interpretation.

Table 1. (Continued.)			
Characteristic	Semaglutide (N = 1767)	Placebo (N = 1766)	Total (N = 3533)
Medication use — no. (%)			
SGLT2 inhibitor	277 (15.7)	273 (15.5)	550 (15.6)
ACE inhibitor	625 (35.4)	615 (34.8)	1240 (35.1)
ARB	1066 (60.3)	1061 (60.1)	2127 (60.2)
Lipid-lowering drug	1418 (80.2)	1416 (80.2)	2834 (80.2)
Diuretic agent	870 (49.2)	910 (51.5)	1780 (50.4)
Insulin	1083 (61.3)	1085 (61.4)	2168 (61.4)

Background medication use: SGLT2 inhibitors were used in only 15.6% at baseline; ACEi/ARB use was high, and insulin use was common.

Main efficacy findings

Treatment effects favored semaglutide across kidney, cardiovascular and survival outcomes.

0.76

HR for primary kidney composite

20

3-year NNT for one primary event

+1.16

ml/min/1.73 m²/yr slower eGFR decline

0.80

HR for all-cause death

Primary outcome: 331/1767 events with semaglutide vs 410/1766 with placebo; HR 0.76, 95% CI 0.66-0.88, P=0.0003.

Interpretation: relative effect is strong, but absolute benefit is driven by high baseline risk. This is relevant to tertiary medicine wards with high-risk diabetic CKD patients.

Figure 1: primary and confirmatory secondary outcomes

Cumulative incidence curves and eGFR slope.

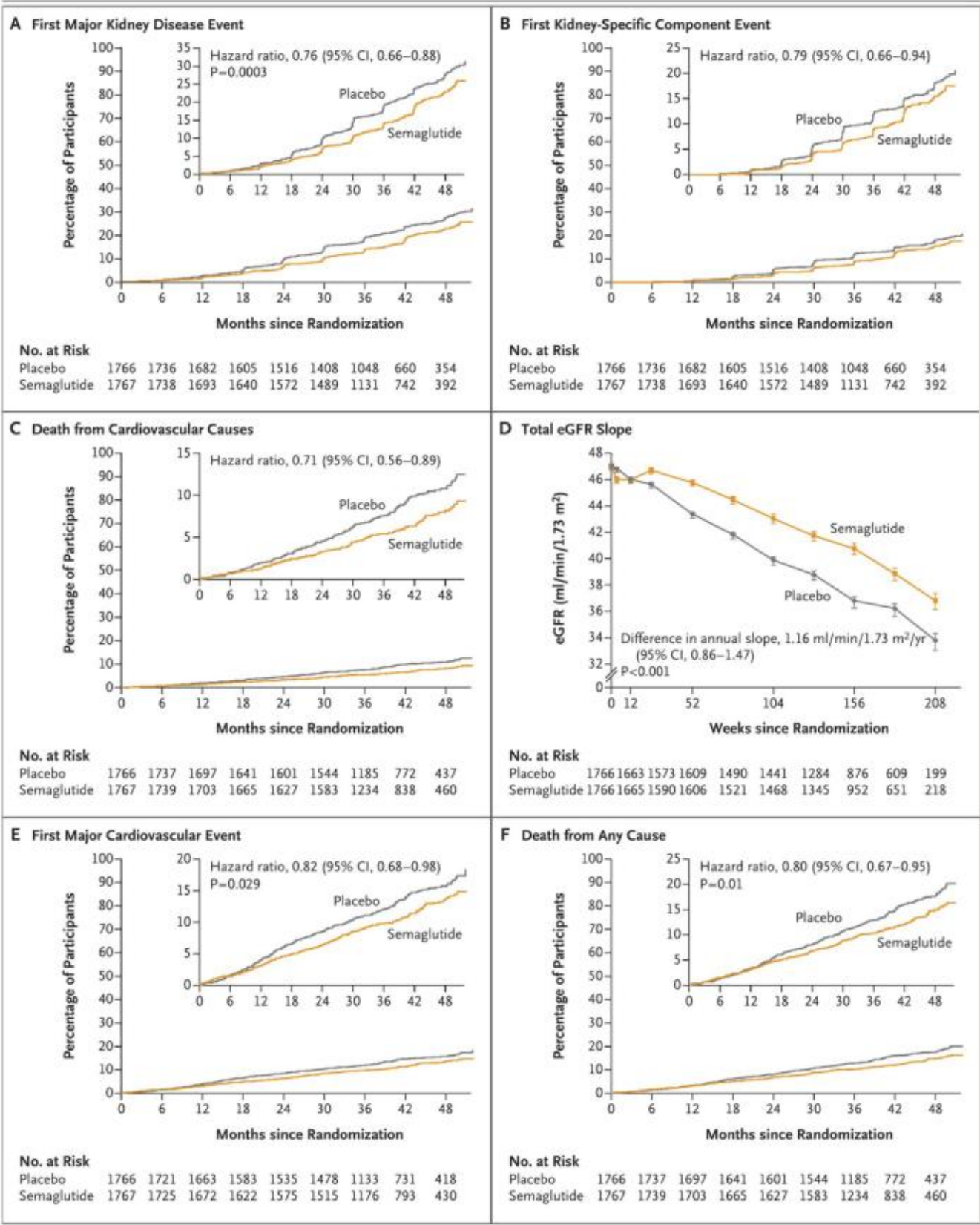
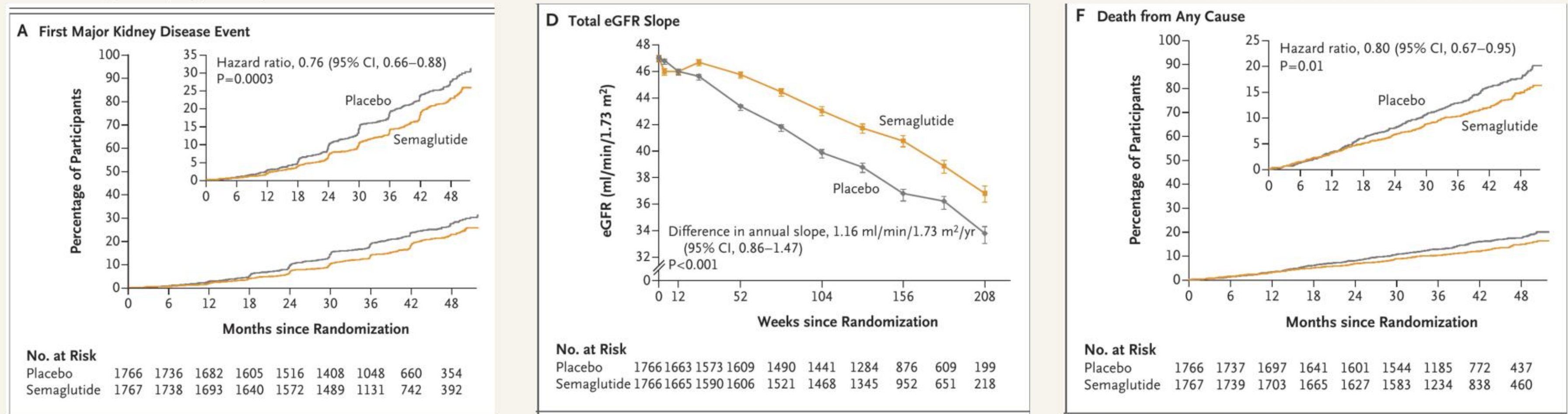


Figure 1: what separates clinically?

The curves separate progressively over time.



- Primary kidney composite: clear difference by 3-4 years.
- eGFR slope: persistent slower decline with semaglutide.
- Mortality and MACE benefits were statistically significant but closer to the threshold.

Table 2: primary and confirmatory outcomes

Expanded crop; all row labels and columns kept visible.

Table 2. Efficacy and Safety Outcomes.*					
Outcome	Semaglutide (N=1767)	Placebo (N=1766)	Hazard Ratio (95% CI)	Estimated Difference (95% CI)	P Value
Primary outcome: major kidney disease events — no. (%)†	331 (18.7)	410 (23.2)	0.76 (0.66 to 0.88)	—	0.0003
Components of primary outcome — no. (%)					
Persistent ≥50% reduction from baseline in eGFR	165 (9.3)	213 (12.1)	0.73 (0.59 to 0.89)	—	—
Persistent eGFR <15 ml/min/1.73 m ²	92 (5.2)	110 (6.2)	0.80 (0.61 to 1.06)	—	—
Initiation of kidney-replacement therapy	87 (4.9)	100 (5.7)	0.84 (0.63 to 1.12)	—	—
Death from kidney-related causes	5 (0.3)	5 (0.3)	0.97 (0.27 to 3.49)	—	—
Death from cardiovascular causes	123 (7.0)	169 (9.6)	0.71 (0.56 to 0.89)	—	—
Composite of kidney-specific components of the primary outcome	218 (12.3)	260 (14.7)	0.79 (0.66 to 0.94)	—	—
Confirmatory secondary outcomes					
Mean annual rate of change in eGFR — ml/min/1.73 m ²	−2.19	−3.36	—	1.16 (0.86 to 1.47)	<0.001
Major cardiovascular events — no. (%)	212 (12.0)	254 (14.4)	0.82 (0.68 to 0.98)	—	0.029
Death from cardiovascular causes	123 (7.0)	169 (9.6)	0.71 (0.56 to 0.89)	—	—
Nonfatal myocardial infarction	52 (2.9)	64 (3.6)	0.80 (0.55 to 1.15)	—	—
Nonfatal stroke	63 (3.6)	51 (2.9)	1.22 (0.84 to 1.77)	—	—
Death from any cause — no. (%)	227 (12.8)	279 (15.8)	0.80 (0.67 to 0.95)	—	0.01

Table 2 continued: supportive and other outcomes

Expanded crop; no table text is cut off.

Supportive secondary outcomes					
Ratio of urinary albumin-to-creatinine ratio at week 104 to urinary albumin-to-creatinine ratio at baseline	0.60	0.88	0.68 (0.62 to 0.75)‡	—	—
Mean change in body weight from baseline to week 104 — kg	−5.55	−1.45	—	−4.10 (−4.56 to −3.65)	—
Mean change in glycated hemoglobin level from baseline to week 104 — percentage points	−0.87	−0.06	—	−0.81 (−0.90 to −0.72)	—
Mean change in systolic blood pressure from baseline to week 104 — mm Hg	−3.79	−1.55	—	−2.23 (−3.33 to −1.13)	—
Mean change in diastolic blood pressure from baseline to week 104 — mm Hg	−0.23	−1.01	—	0.78 (0.16 to 1.41)	—
Mean change in eGFR from baseline to week 12 — ml/min/1.73 m ²	−1.07	−1.05	—	−0.03 (−0.56 to 0.51)	—
Mean annual rate of change in eGFR from week 12 to end of trial — ml/min/1.73 m ²	−2.36	−3.30	—	0.94 (0.62 to 1.26)	—
Mean change in eGFR by the cystatin C equation from baseline to week 104 — ml/min/1.73 m ²	−2.01	−5.41	—	3.39 (2.63 to 4.15)	—
Major adverse limb event in a time-to-first-event analysis — no. of events	16	28	0.56 (0.30 to 1.02)	—	—
No. of severe hypoglycemic episodes	47	46	1.02 (0.62 to 1.67)‡	—	—
Supplementary analysis: death from noncardiovascular and non-kidney-related causes in a time-to-first-event analysis — no. of events	99	105	0.93 (0.70 to 1.22)	—	—

Safety findings

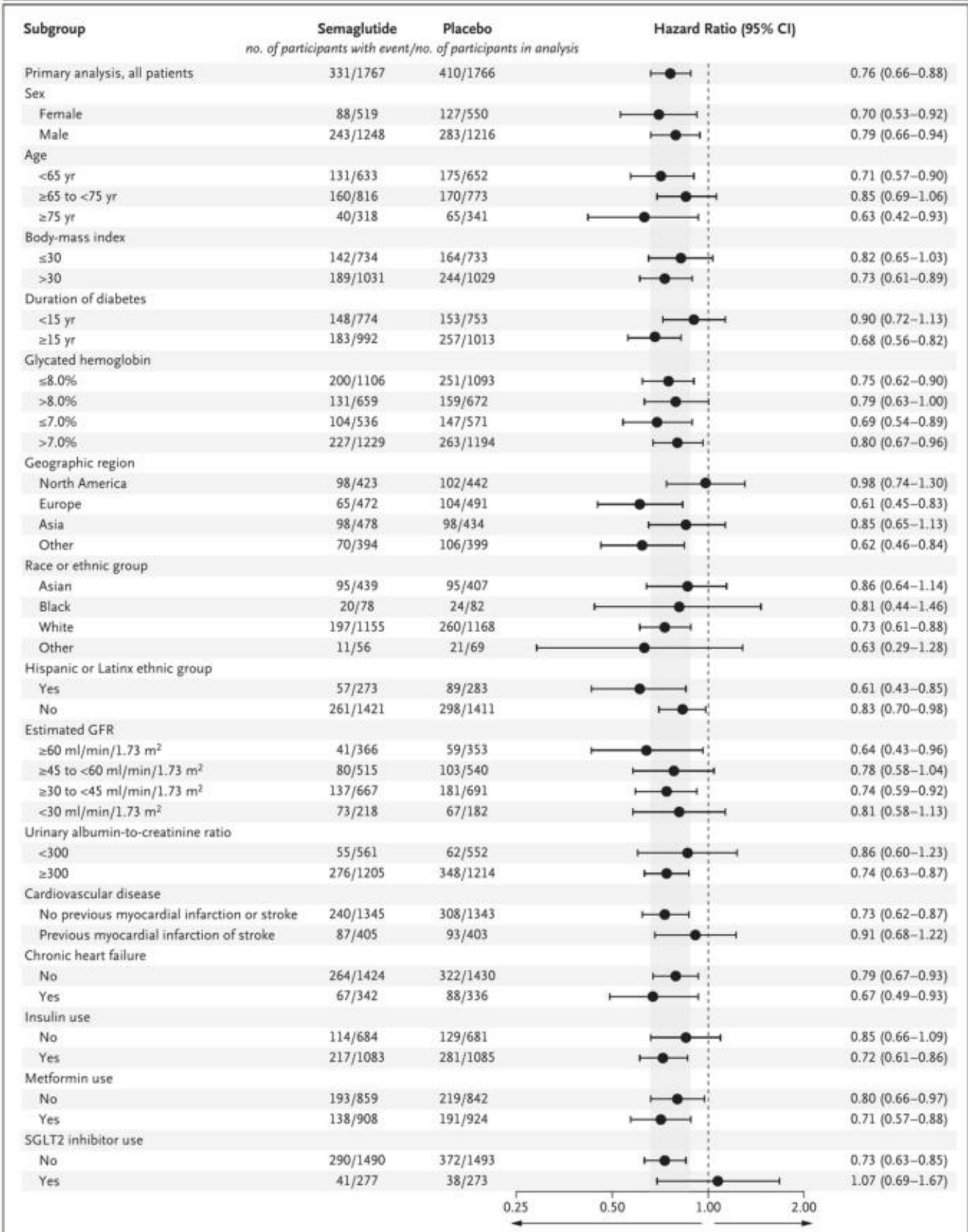
Overall serious adverse events were fewer, but discontinuation was slightly higher with semaglutide.

Table 3. Safety Outcomes.		
Adverse Event	Semaglutide (N = 1767)	Placebo (N = 1766)
	no. of participants (%)	
Serious adverse event	877 (49.6)	950 (53.8)
Adverse event leading to permanent discontinuation of semaglutide or placebo	233 (13.2)	211 (11.9)
Prespecified adverse events of special interest		
Diabetic retinopathy*	402 (22.8)	398 (22.5)
Covid-19–related disorder	358 (20.3)	404 (22.9)
Serious adverse event: cardiovascular disorder	273 (15.4)	319 (18.1)
Heart failure*	133 (7.5)	175 (9.9)
Acute kidney failure*	172 (9.7)	182 (10.3)
Malignant tumor*	120 (6.8)	104 (5.9)
Serious adverse event: gastrointestinal disorder	95 (5.4)	94 (5.3)
Serious adverse event: rare event	48 (2.7)	57 (3.2)
Acute gallbladder disease*	32 (1.8)	39 (2.2)
Severe hypoglycemia*	37 (2.1)	37 (2.1)
Medication error*	15 (0.8)	13 (0.7)
Serious adverse event: hepatic disorder	18 (1.0)	20 (1.1)
Acute pancreatitis*	10 (0.6)	7 (0.4)
Serious adverse event: allergic reaction	6 (0.3)	9 (0.5)
Serious adverse event: abuse and misuse	1 (0.1)	4 (0.2)
Serious adverse event: suspected transmission of infectious agent through semaglutide or placebo	0	1 (0.1)

- Serious adverse events: 49.6% vs 53.8%.
- Permanent discontinuation: 13.2% vs 11.9%.
- GI disorders were the main driver of discontinuation: 4.5% vs 1.1%.
- Diabetic retinopathy events were similar; serious eye disorders were numerically higher.

Figure 2: subgroup analysis

Most subgroup estimates were directionally consistent, but not all were precise.



Subgroup interpretation

Useful clinically, but should not be over-read.

What supports consistency

- Overall HR 0.76 remained the central estimate.
- Effect favored semaglutide in most clinically relevant groups.
- Benefit seen in Asian subgroup estimate, though not India-specific.

Where precision is limited

- SGLT2 inhibitor subgroup: HR 1.07, wide CI 0.69-1.67.
- Black and “other” race subgroups had few events.
- Stroke component numerically favored placebo but CI crossed 1.

Critical appraisal: forest plots are hypothesis-generating unless powered and adjusted for interaction testing.

CASP: Results appraisal

Size and precision of benefit.

Similar at baseline?

Yes - Table 1 supports balance.

Treatment effect size?

24% relative risk reduction for primary composite; NNT 20 over 3 years.

Precision?

Primary outcome CI 0.66-0.88: reasonably precise. Some secondary/subgroup CIs are wider.

Clinical importance?

High in selected high-risk CKD patients because kidney failure, CV death and all-cause death matter.

Equal treatment?

Mostly yes, but evolving background care limits modern combination interpretation.

DISCUSSION

Main findings from the discussion

Semaglutide becomes a serious kidney-protective option in T2D + CKD.

- Semaglutide reduced major kidney disease events by 24%.
- It also reduced major cardiovascular events and all-cause mortality.
- eGFR decline slowed by 1.16 ml/min/1.73 m²/year.
- Mechanism is likely multifactorial: weight, glycemia, BP, inflammation, oxidative stress and fibrosis.
- The study supports semaglutide as an add-on to standard CKD risk-reduction therapy.

Bottom line for teaching: FLOW changes GLP-1 RA discussion from “albuminuria surrogate benefit” to “hard kidney and survival outcome trial evidence.”

Comparison with literature since FLOW

Later analyses generally support the main result, but combination sequencing remains open.

- Prespecified FLOW analyses reported CV benefit consistency across CKD severity categories.
- Nature Medicine FLOW analysis: benefits appeared present with and without baseline SGLT2 inhibitor use, but baseline SGLT2 users were few.
- A 2025 Lancet Diabetes & Endocrinology meta-analysis including FLOW found GLP-1 RAs reduce clinically important kidney and cardiovascular outcomes.
- US FDA expanded semaglutide/Ozempic indication in 2025 for diabetic CKD risk reduction.

Sources searched: Nature Medicine 2024 FLOW SGLT2 analysis; European Heart Journal 2025 CKD severity analysis; Lancet Diabetes & Endocrinology 2025 meta-analysis; Reuters/FDA label update 2025.

Applicability to medicine patients in North Indian medical college, Ghaziabad

High clinical relevance, but with access and monitoring constraints.

Highly applicable when patient resembles FLOW

- T2D with CKD stage G3-G4 and albuminuria.
- Already on ACEi/ARB unless contraindicated.
- High CV risk, obesity, insulin use or long diabetes duration.
- Reliable follow-up for renal function, GI adverse effects and retinopathy.

Less directly applicable / practical barriers

- Patients without albuminuria or low-risk CKD were not the main trial population.
- Many local patients may not afford long-term injectable semaglutide.
- Need to prioritize SGLT2 inhibitor and ACEi/ARB foundation where possible.
- GI intolerance, weight loss, dehydration risk and retinopathy history need review.

Local evidence context: India has a large diabetes burden and diabetic kidney disease is common among T2D patients; cost and availability are central implementation issues.

Practical medicine-ward approach

A simple checklist for diabetic CKD patients.

- 1 Confirm FLOW-like risk** eGFR 25-75 + albuminuria; exclude non-diabetic active renal disease if suspected.
- 2 Optimize foundation** ACEi/ARB, BP, statin, SGLT2 inhibitor if eligible, glycemic plan.
- 3 Consider semaglutide** Especially obesity, high CV risk, persistent albuminuria, high residual risk.
- 4 Monitor safely** GI intolerance, volume depletion, renal function, retinopathy, affordability/adherence.

Do not use this trial to justify semaglutide as a substitute for basic CKD care. Use it as an additional risk-reduction tool.

Final critical appraisal verdict

High-quality evidence with important implementation caveats.

High

Internal validity

Moderate

Generalizability to Indian practice

Strong

Clinical importance

- FLOW is a practice-changing RCT for high-risk T2D + CKD.
- Benefits include kidney, cardiovascular and mortality outcomes.
- The biggest caution is current standard therapy context: SGLT2 and finerenone uptake was modest in the trial.
- For North Indian medical college patients, the key issue is choosing the right high-risk patient and sustaining therapy.

Key sources used

Primary PDF plus targeted literature search for critical appraisal.

- Perkovic V et al. Effects of semaglutide on chronic kidney disease in patients with type 2 diabetes. NEJM 2024.
- Mann JFE et al. Effects of semaglutide with and without concomitant SGLT2 inhibitor use in FLOW. Nature Medicine 2024.
- Mahaffey KW et al. Cardiovascular outcomes with semaglutide by CKD severity in FLOW. European Heart Journal 2025.
- Badve SV et al. GLP-1 receptor agonists on kidney and cardiovascular outcomes: meta-analysis. Lancet Diabetes & Endocrinology 2025.
- ADA Standards of Care 2026 CKD chapter; KDIGO diabetes/CKD guidance; FDA/Reuters label update 2025.
- ICMR-INDIAB / Lancet Diabetes & Endocrinology 2023; Indian DKD prevalence meta-analysis 2025.

No outside figures were used. Tables and figures were cropped from the study PDF.